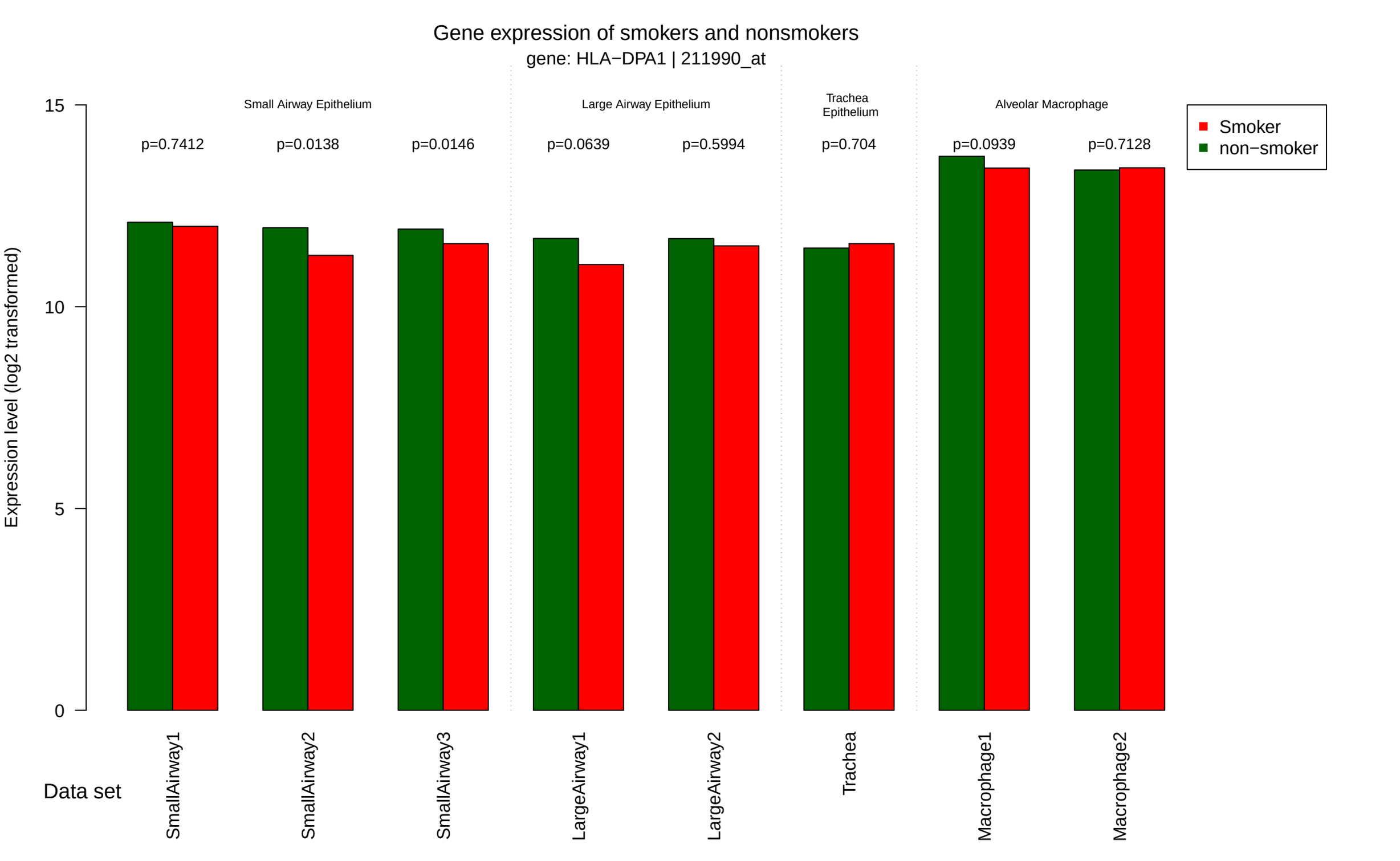
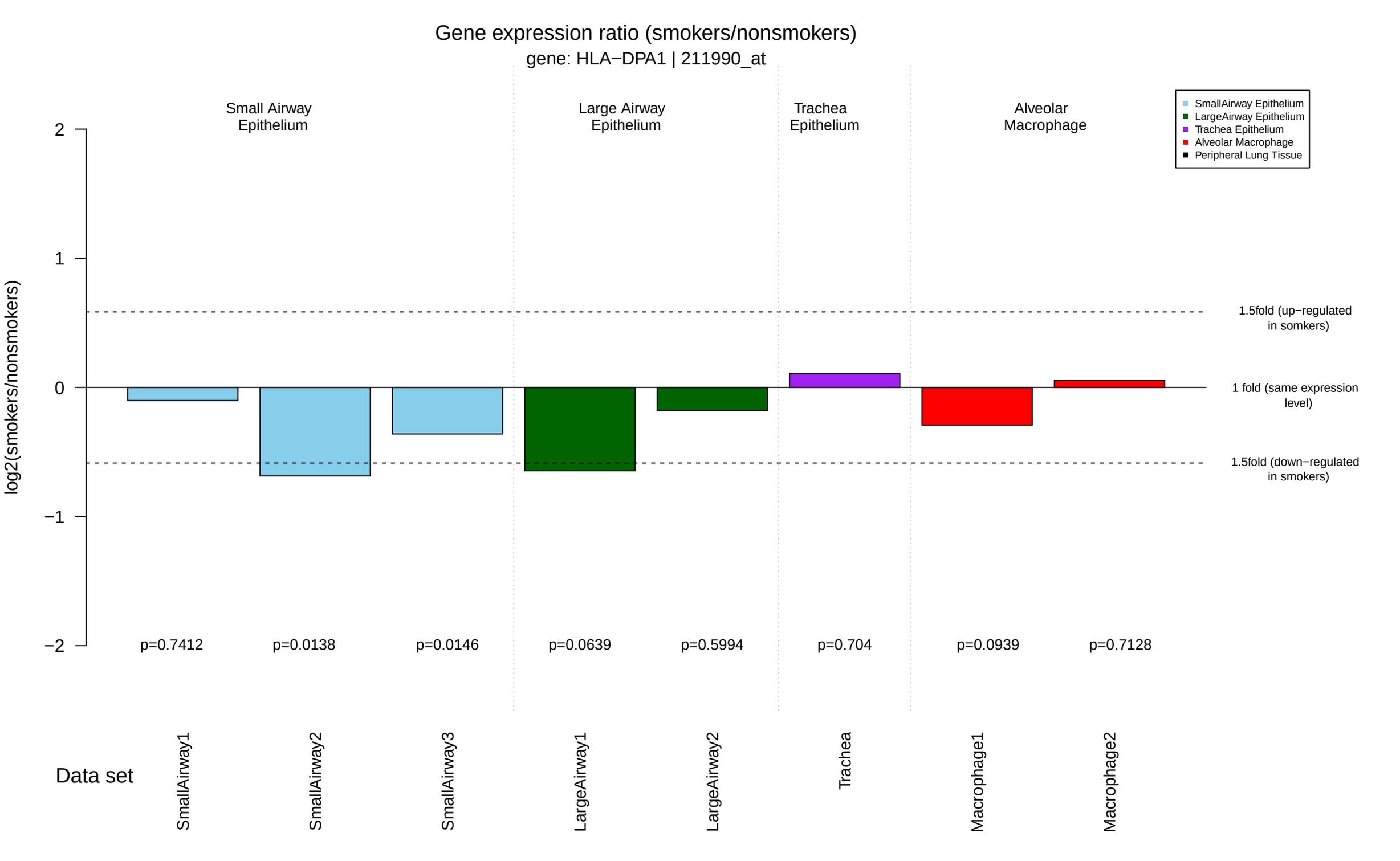
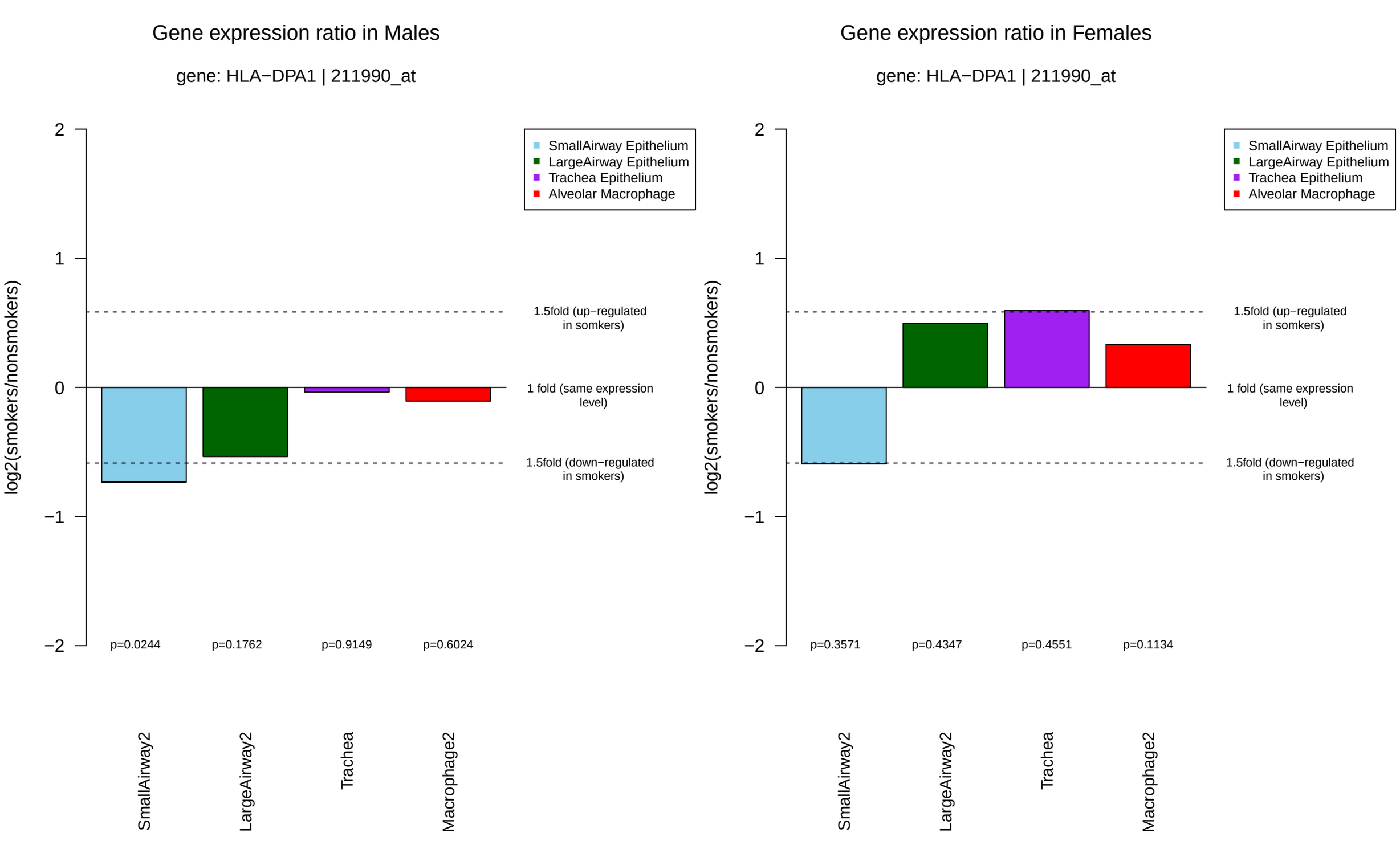
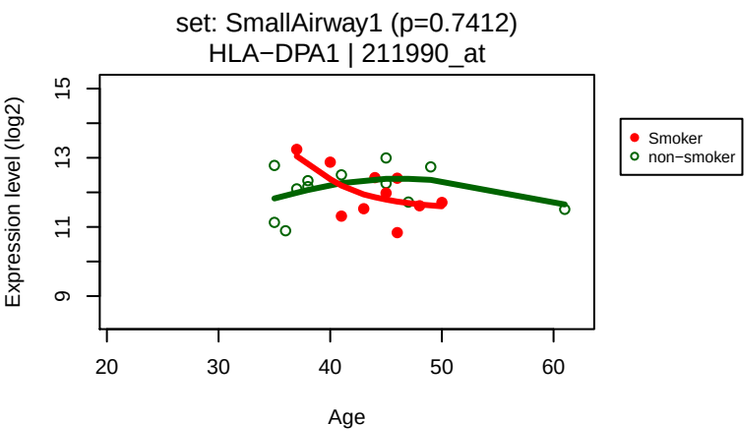
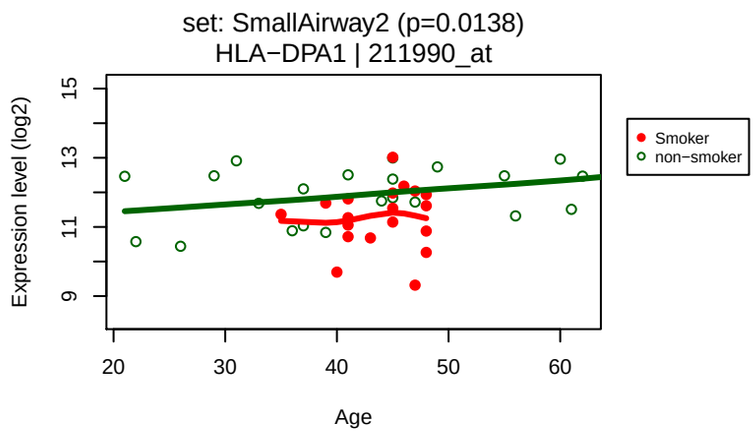
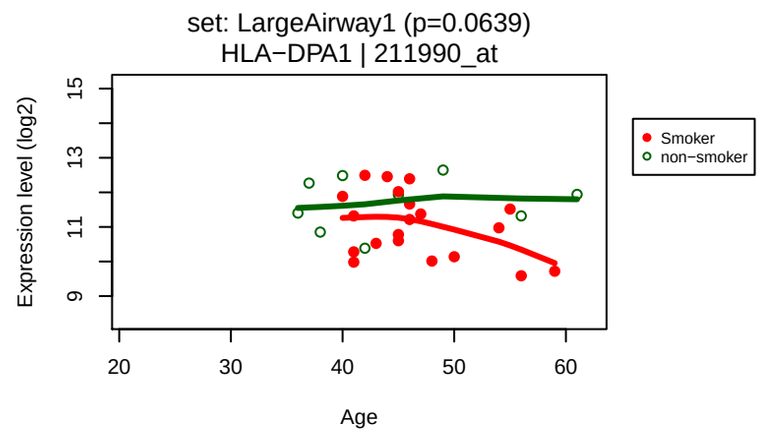

Gene expression of smokers and nonsmokers
gene: HLA−DPA1 | 211990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7412
p=0.0138
p=0.0146
p=0.0639
p=0.5994
p=0.704
p=0.0939
p=0.7128
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

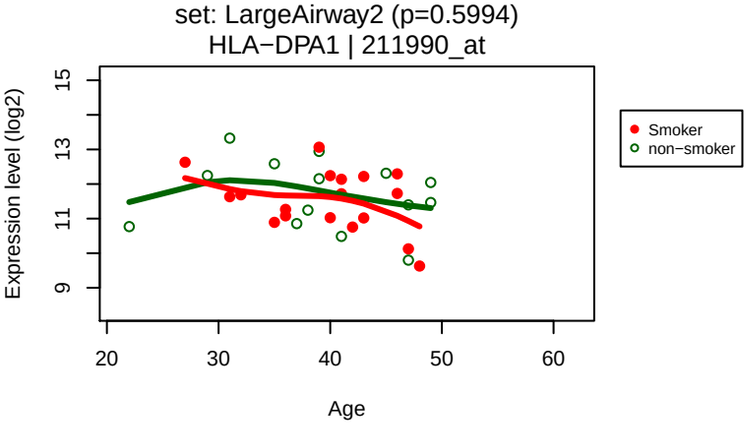

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DPA1 | 211990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7412
p=0.0138
p=0.0146
p=0.0639
p=0.5994
p=0.704
p=0.0939
p=0.7128
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

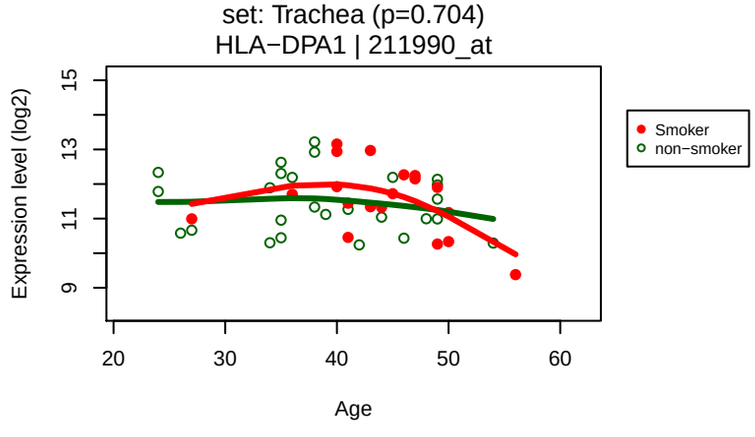

Gene expression ratio in Males
gene: HLA−DPA1 | 211990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0244
p=0.1762
p=0.9149
p=0.6024
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DPA1 | 211990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3571
p=0.4347
p=0.4551
p=0.1134
SmallAirway2
LargeAirway2
Trachea
Macrophage2

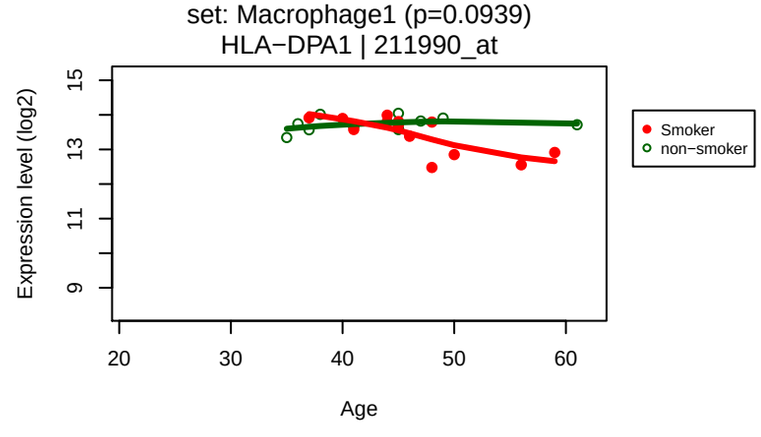

set: SmallAirway1 (p=0.7412)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker

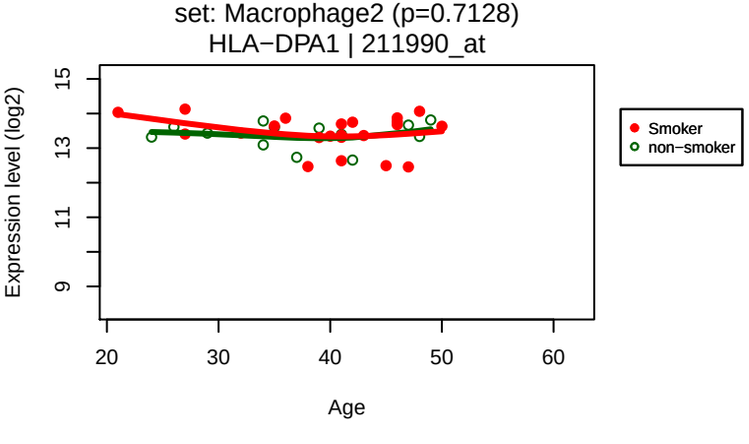

set: SmallAirway2 (p=0.0138)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker

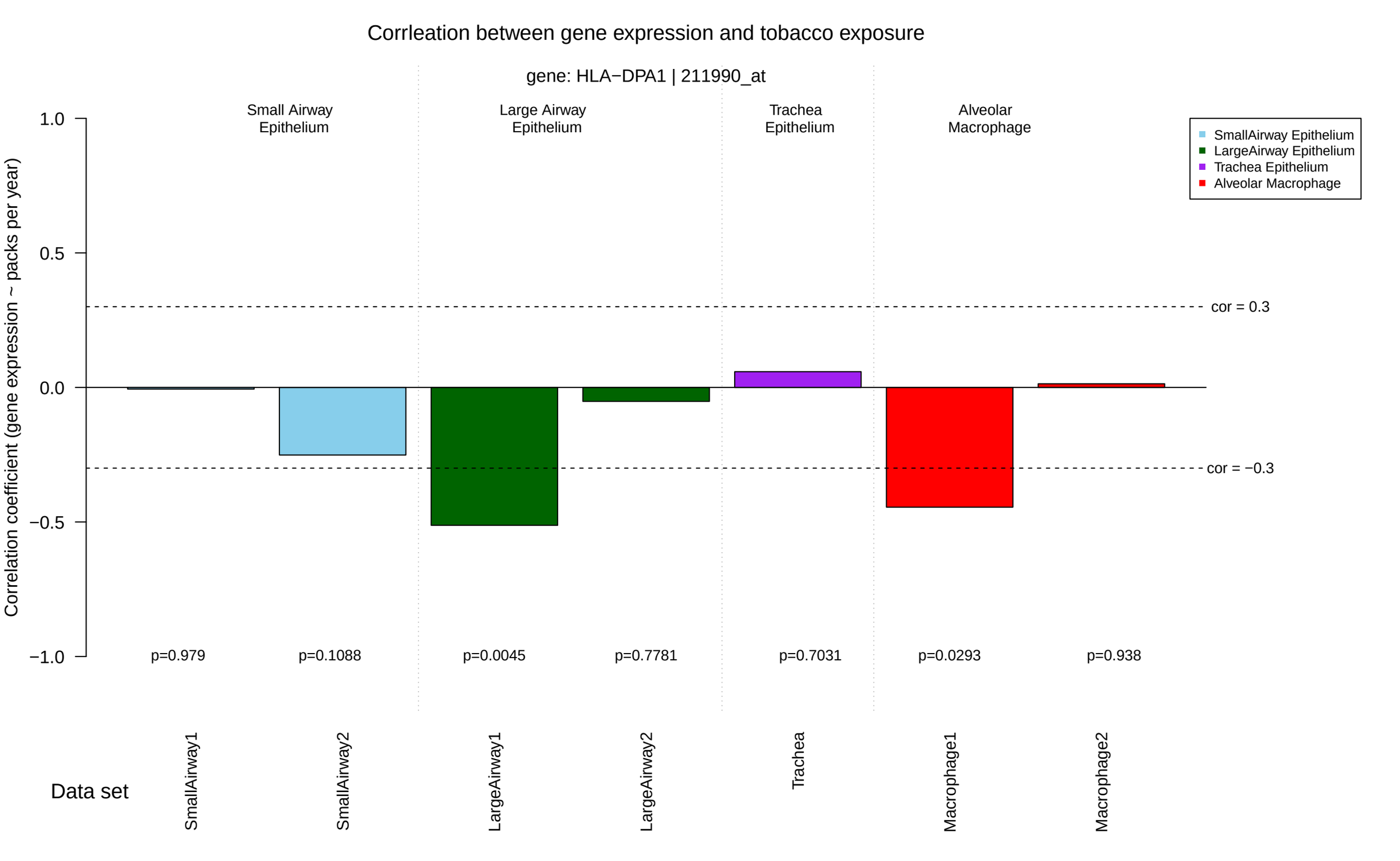

set: LargeAirway1 (p=0.0639)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5994)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.704)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0939)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7128)
HLA−DPA1 | 211990_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DPA1 | 211990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.979
p=0.1088
p=0.0045
p=0.7781
p=0.7031
p=0.0293
p=0.938
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2